Tre modalità a confronto: pipeline, chiamata singola, librerie

Qwen3.8-27B, Gemma 4 31B e Mistral Small 4 (AWQ 4-bit) con il prompt v2 su sorgenti PDF e LaTeX · run su CINECA Leonardo del 27/10/2026 (job 62440414, 62440416, 62442769) · **aggiornato il 29/10/2026** con le correzioni della valutazione, il gold set delle figure e il prompt v2.1 (job 62571383, 62571407–62571450) · Riccardo Aldrovandi

Cosa è cambiato rispetto alla prima versione del 27/10.

- **L2 sul LaTeX:** era sottostimato per un difetto della valutazione. I rimandi "Sec. 3.2" / "Eq. (11)", che nel sorgente non esistono, e i numeri attaccati ai comandi LaTeX (`\sim0.14`, `Smerald2014`) risultavano non ancorati. Dopo la correzione il divario PDF/LaTeX quasi sparisce: la conclusione "il LaTeX è peggio" era sbagliata.
- **Valori letti dai grafici:** il controllo dei numeri penalizzava i valori letti a occhio e dichiarati "approx., read from the figure", che per costruzione non sono nel testo. Ora sono esclusi. Questo cambia soprattutto Qwen in single, che legge le figure meglio di tutti.
- **Transformer-XL:** mancava per Qwen sul LaTeX (figure PNG da 10000×2000 pixel). Ora c'è: ho ridimensionato le figure e rilanciato le 6 analisi; 5 sono riuscite, 1 è stata rifiutata perché ha superato il limite di token.
- **Nuova metrica, i controlli delle figure:** 42 domande verificabili su 14 figure dei 10 paper (asintoti, picchi, quale curva decade prima…). Colma il punto cieco segnalato nell'appendice A. Con questa metrica il giudizio su Qwen cambia: la modalità single, che avevo indicato come la peggiore, è quella che legge meglio le figure.
- **Caratteri corrotti di Mistral:** la causa è confermata (`skip_special_tokens=false`) ed è corretta.
- **Nuova sezione 7:** il prompt v2.1 e la modalità ibrida, con la scelta per il set di fine-tuning.

1. Cosa abbiamo misurato

Ogni modello ha analizzato i 10 paper del gold set: 6 di fisica e 4 di ML, di cui 9 con il sorgente LaTeX (anti-k_t non lo ha). Ha usato le tre modalità e, per ciascuna, 3 seed diversi: in tutto 171 analisi per modello e 508 analisi valide in totale (503 nella prima versione, più le 5 di Transformer-XL rilanciate per Qwen). Lo schema di output è lo stesso per tutti: il JSON v2, con figure, equazioni, risultati e così via in campi separati.

Modalità	Cosa riceve il modello	Chiamate al modello
A · pipeline	Il testo del paper, più i dati che il modello stesso estrae da ogni figura con la visione. Su quei dati PySR cerca una formula, che viene aggiunta al prompt.	una per figura, più quella finale
B · single	Il testo, più le immagini delle pagine con figure (dal PDF) o i file delle figure (dal LaTeX), tutto in un'unica chiamata, come si farebbe caricando il paper in una chat. Nessuna analisi delle figure da parte nostra.	una sola
C · libs	Il testo, più i dati dei grafici letti solo con librerie: tracciati vettoriali del PDF calibrati sulle tacche degli assi, coordinate pgfplots, tabelle. PySR come in A.	solo quella finale

Le metriche.

- **Recall**: la quota pesata dei fatti attesi (dalle rubriche dei 10 paper) che compaiono nell'output. Misura quante informazioni il modello estrae.
- **L2**: misura se ciò che il modello scrive è ancorato al paper. È la media di tre controlli: i numeri citati, i riferimenti a figure, tabelle ed equazioni, e i dataset nominati. Nelle modalità A e C c'è anche una componente **"visual"**, cioè la quota delle formule PySR fornite che il modello riporta, inquadrate come approssimazioni.
- Gli intervalli di confidenza sono bootstrap al 95% sui paper. Dico che una differenza è "separabile" quando l'intervallo della differenza tra due configurazioni, calcolata sugli stessi paper, non contiene lo zero.

> **Il L2 complessivo non è confrontabile tra le modalità.** La modalità single non riceve formule, quindi non ha la componente "visual", che è proprio quella su cui i modelli differiscono di più. Nel confronto uso quindi il **L2 senza la componente visual** e tratto l'uso delle formule come metrica separata.

2. La risposta breve

> **Sulla quantità di informazioni estratte, le tre modalità quasi non si distinguono, per nessuno dei tre modelli.** Dentro uno stesso modello la recall varia di 1–2 punti, al limite del rumore. Il motivo è che quasi tutti i fatti delle rubriche (risultati, metodi, approssimazioni, valori dichiarati) stanno nel testo, che arriva identico in tutte e tre le modalità. Le figure aggiungono soprattutto le formule ricavate dai grafici, che le rubriche premiano poco.
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> Le modalità si distinguono invece su due aspetti:
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> 1. quanto il modello **usa i dati delle figure** che gli passiamo;
> 2. quanto restano **ancorati al paper** i numeri e i dataset che scrive.
>
> Su questi due punti i tre modelli si comportano in modo opposto, quindi la modalità migliore dipende dal modello. **Aggiornamento 29/10:** c'è un terzo aspetto, misurato ora dai controlli delle figure: quanto bene il modello **legge i grafici**. Su questo vince la chiamata singola di Qwen, e con il prompt v2.1 migliorano tutti (sezione 7).

3. I numeri (corretti il 29/10)

Modello	Sorgente	Modalità	Recall	L2 senza visual	Formule riportate / fornite	Controlli figure
Qwen3.8	PDF	pipeline	0.979	0.982	77% (491/638)	43%
		single	0.992	0.963	–	60%
		libs	0.979	0.981	84% (248/297)	37%
	LaTeX	pipeline	0.980	0.966	95% (560/591)	52%
		single	0.983	0.908	–	59%
		libs	0.982	0.965	76% (188/246)	32%
Gemma 4	PDF	pipeline	0.929	0.981	25% (171/696)	24%
		single	0.927	0.984	–	27%
		libs	0.930	0.984	40% (116/293)	28%
	LaTeX	pipeline	0.917	0.956	17% (108/628)	30%
		single	0.934	0.944	–	30%
		libs	0.912	0.960	37% (90/246)	27%
Mistral Small 4	PDF	pipeline	0.966	0.939	7% (22/302)	33%
		single	0.948	0.979	–	38%
		libs	0.959	0.948	3% (9/297)	40%
	LaTeX	pipeline	0.959	0.829	8% (21/279)	31%
		single	0.949	0.902	–	34%
		libs	0.943	0.893	3% (7/246)	30%

Recall: media sui record dei 3 seed. L2 senza visual: media dei controlli su numeri, riferimenti e dataset, con le correzioni del 29/10 (LaTeX e valori letti dai grafici). Formule: totale sui 3 seed; ce ne sono di più in pipeline che in libs perché le librerie leggono solo i grafici vettoriali con assi numerici. Controlli figure: quota delle 42 domande sulle figure a cui l'output risponde correttamente (gold/figure_checks.jsonl).

Fra i modelli la classifica sulla recall è netta e separabile in quasi tutti i confronti: **Qwen (≈0.98) > Mistral (≈0.95) > Gemma (≈0.93)**. Fra le modalità, dentro lo stesso modello, la recall non è separabile, con un'unica eccezione minima per Mistral. **Sui controlli delle figure** la classifica è la stessa, ma con distacchi molto più grandi: Qwen in single arriva al 60%, mentre gli altri restano tra il 24% e il 40% in tutte le modalità.

4. Modello per modello

Qwen3.8: single legge meglio le figure, pipeline e libs riportano le formule

- **È l'unico modello che usa davvero le formule.** Riporta il 76–95% di quelle che riceve, quasi sempre come approssimazioni dichiarate. Nelle modalità A e C l'output contiene quindi una descrizione quantitativa delle curve che in single non c'è.

- **Ma in single legge meglio le figure stesse:** 60% dei controlli sul PDF, contro il 43% della pipeline e il 37% di libs; sul LaTeX 59%, contro 52% e 32%. Guardando l'immagine coglie asintoti, picchi e confronti tra curve che le formule non dicono (su Flux tube la formula PySR va a zero, mentre la curva vera si stabilizza a 0.6).
- **La recall è la stessa** nelle tre modalità (0.98–0.99). Il vantaggio di single sul PDF, 0.992 contro 0.979, è separabile ma piccolo.
- **Ancoraggio:** dopo le correzioni single resta un po' sotto (0.963 sul PDF, 0.908 sul LaTeX, contro circa 0.97–0.98). Il motivo è che a volte scrive valori letti dal grafico senza marcarli come approssimati, cosa che il prompt chiede.

Migliore: single per la lettura delle figure e la completezza; pipeline o libs se servono le formule (libs costa meno, a parità di risultato). Nella prima versione avevo indicato single come la peggiore: era un effetto della metrica, che non vedeva la lettura delle figure e penalizzava i valori letti dai grafici.

Gemma 4: single la migliore, pipeline la peggiore

- **La recall è piatta**, ≈0.93 in tutte e tre le modalità; anche i controlli delle figure sono piatti e bassi (24–30%).
- **Usa poco le formule**: il 17–25% in pipeline e il 37–40% in libs.
- **La pipeline è la peggiore.** La digitalizzazione dei grafici fatta da Gemma stessa produce molte formule poco affidabili (sul PDF circa 450 a bassa confidenza sui 3 seed, contro circa 210 in libs), che poi Gemma scarta quasi tutte.
- **Single** dà la stessa recall con una sola chiamata e numeri ben ancorati (0.984 sul PDF).

Migliore: single. Peggiore: pipeline. Invariato rispetto alla prima versione.

Mistral Small 4: pipeline la migliore sulla recall, ma conviene single

- **Ignora quasi del tutto le formule**: ne riporta il 3–8%. Inoltre la sua digitalizzazione è la più debole: produce più del doppio delle curve "segnaposto" (punti inventati) degli altri modelli (sul LaTeX, seed 0: 107 contro 48 di Qwen e 44 di Gemma).
- **La pipeline** batte single sulla recall in modo appena separabile: sul PDF 0.966 contro 0.948, intervallo [+0.001, +0.039].
- **Il materiale extra peggiora l'ancoraggio:** in pipeline e libs nomina dataset che non compaiono nel paper (controllo dei dataset 0.77–0.86, contro 1.00 in single).
- **Controlli figure** tra il 31% e il 40% in tutte le modalità: nessuna legge bene i grafici.

Migliore: single per affidabilità. Peggiore: libs. Invariato rispetto alla prima versione.

5. Quadro complessivo (prompt v2)

Modello	Migliore	Peggiore	Cosa decide
Qwen3.8	single (figure, recall); libs per le formule	– (dipende dall'obiettivo)	legge davvero le immagini e usa anche le formule
Gemma 4	single	pipeline	digitalizzazione rumorosa e poco usata
Mistral Small 4	single	libs	ignora le formule; il materiale extra peggiora l'ancoraggio

1. **Con il prompt v2 la lettura delle figure la fa bene solo Qwen**, e la fa meglio guardando l'immagine (single) che leggendo le formule (pipeline, libs). Per Gemma e Mistral la chiamata singola dà gli stessi fatti a un costo molto minore.
2. **La lettura dei grafici con le librerie (C) vale quanto la digitalizzazione fatta dal modello (A)** per recall e formule, nei grafici vettoriali con assi numerici: costa molto meno ed è deterministica. Il limite è la copertura, perché non legge le figure raster.
3. **PDF e LaTeX sono equivalenti:** stessa recall, e dopo la correzione della valutazione anche L2 quasi uguale (per esempio Qwen libs 0.981 sul PDF contro 0.965 sul LaTeX; prima era 0.901). Quello che resta sono valori calcolati dal modello e un ancoraggio più debole di Mistral sul LaTeX.

6. Limiti

- **Campione piccolo**: 10 paper (9 per il LaTeX). Le differenze di 1–2 punti sono vicine al rumore, e i confronti sul LaTeX sono indicativi.
- **5 analisi rifiutate su 513** (3 di Gemma, 2 di Qwen, tra cui una di Transformer-XL rilanciata), tutte per aver superato i 32.768 token di output, una volta con un loop di ripetizione di Gemma. Mistral non ne ha nessuna.
- **I controlli delle figure sono una bozza da rivedere:** espressioni regolari scritte da me su 14 figure, tarate a mano su Flux tube. Possono ancora dare qualche falso positivo o negativo; le tendenze tra modelli però sono ampie (60% contro 24–40%).
- **Le rubriche misurano i fatti, non la correttezza fine:** una frase sbagliata che contiene le parole chiave giuste conta come trovata. I controlli delle figure coprono questo buco solo per le figure.

7. Aggiornamento: prompt v2.1 e modalità ibrida (solo PDF)

Dopo la prima versione di questo documento ho preparato il **prompt v2.1**, pensato per i difetti visti sopra:

- per ogni figura un campo `curves`: valore iniziale, massimo o minimo, valore finale o asintoto e andamento di ogni curva;
- limiti e dataset solo se scritti dagli autori;
- il campo delle formule PySR lo compila il codice, non più il modello;
- nella chiamata singola anche i ritagli delle figure, oltre alle pagine.

In più c'è una **modalità ibrida** (le immagini della chiamata singola più i dati letti dalle librerie), e ho provato una seconda chiamata che verifica le figure contro le immagini. Ogni configurazione ha girato sui 10 paper in PDF con 3 seed, con e senza ragionamento.

Configurazione	Controlli figure	Solo domande che richiedono l'immagine	Recall	L2 senza visual
Qwen v2 single (riferimento)	60%	36%	0.992	0.963
Qwen v2.1 hybrid, con ragionamento	77%	64%	0.992	0.932
Qwen v2.1 single, con ragionamento	77%	61%	0.981	0.940
Qwen v2.1 pipeline / libs	66% / 63%	38% / 31%	0.984 / 0.976	0.961 / 0.970
Qwen v2.1 hybrid, senza ragionamento	68%	58%	0.944	0.975
Mistral v2.1 hybrid, con ragionamento	62%	42%	0.934	0.966
Mistral v2 single (riferimento)	38%	14%	0.948	0.979
Gemma v2.1 hybrid, con ragionamento	52%	36%	0.908	0.980
Gemma v2 single (riferimento)	27%	11%	0.927	0.984

1. **Il campo `curves` migliora la lettura delle figure per tutti e tre i modelli:** Qwen dal 60% al 77%, Mistral da circa il 38% al 62%, Gemma da circa il 27% al 52%. Chiedere valori precisi costringe il modello a guardare il grafico invece di ripetere il testo. Con la v2.1 anche pipeline e libs migliorano (Qwen 66% e 63%), perché il blocco dei dati dei grafici ora riporta anche i punti letti, e non più solo la formula.
2. **Il ragionamento serve:** senza, Qwen perde circa 4–5 punti di recall.
3. **La seconda chiamata di verifica non porta niente:** sposta il punteggio di uno o due punti in entrambe le direzioni (Qwen hybrid 79% prima, 77% dopo; single 76% e 77%). Non conviene pagarla.
4. **L'ancoraggio della v2.1 scende un po' per Qwen** (0.93–0.94): scrive più valori letti dai grafici, e non sempre li marca come approssimati.

Scelta per il set di fine-tuning: Qwen3.8, prompt v2.1, modalità hybrid, con ragionamento e senza la seconda verifica. La recall è la più alta (0.992, come la v2 single) e la lettura delle figure la migliore (77%; 64% sulle domande che richiedono l'immagine).

Appendice A: Flux tube, nove output a confronto

Paper: *Flux tube in turbulent flow and quantum chromodynamics* (sorgente PDF, seed 0). È un paper teorico che propone un'analogia formale fra turbolenza e QCD. Ha una sola figura, che è il caso ideale per vedere come ciascun modello tratta i grafici.

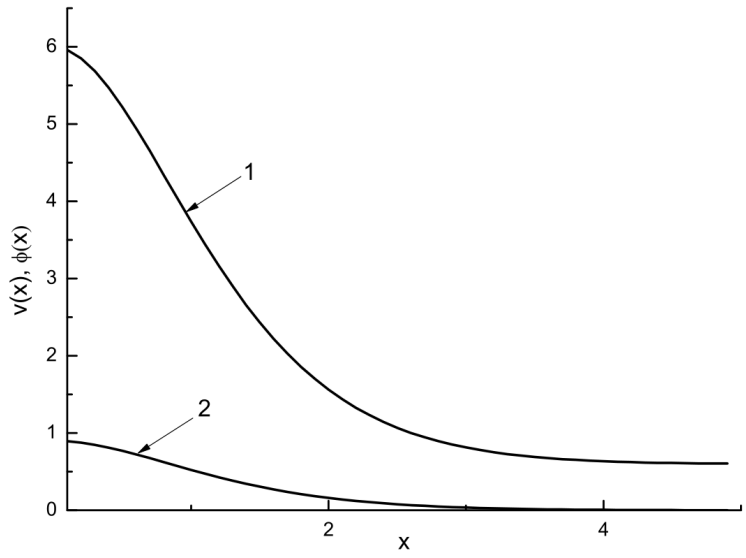


Fig. 1 del paper: profili di v(x) (curva 1) e φ(x) (curva 2), con $\tilde{\alpha} = \tilde{\beta} = -3$ e $\gamma = 6$.

Cosa mostra davvero la figura, che ho verificato a mano:

- v(x) parte da ≈6 e scende, ma **non va a zero**: si stabilizza a ≈0.6. Il testo indica questo come il caso più probabile: "*a laminar fluid $v_\infty \neq 0$ (that is more probably) or in a rest fluid $v_\infty = 0$ (that is less probably)*", e aggiunge che v_∞ va scelto "small enough" perché il flusso esterno sia laminare.
- φ(x) parte da ≈0.9 e va a ≈0 già verso x ≈ 3: decade **più in fretta** di v.

Questi due dettagli separano chi ha letto bene la figura da chi l'ha descritta in modo generico. Dire che v va a zero contraddice la figura e il caso che l'autore considera più probabile, ma non la tesi centrale del paper: il core a tubo e il flusso non nullo, che nell'Eq. 54 è calcolato con $v - v_\infty$, non dipendono dal fatto che v_∞ sia diverso da zero. È quindi un errore di lettura reale, di gravità moderata. Gli output JSON completi sono nell'appendice B.

Riepilogo dei nove output

Modello	Modalità	Recall	Fatti mancanti	Risultati / equazioni	Lettura della Fig. 1
Qwen3.8	pipeline	1.00	–	6 / 8	**corretta**, più 2 formule PySR riportate; $v_\infty \neq 0$ nei risultati
	single	1.00	–	4 / 8	**la più precisa**: legge 6 → ≈0.5 e φ → 0 per x ≈ 2, marcati come approssimati
	libs	0.94	m3 (Heisenberg)	6 / 8	**corretta**, ma qualitativa (libs non ha letto il grafico)
Gemma 4	pipeline	1.00	–	2 / 4	generica ("core pronunciato"), più 2 formule PySR riportate
	single	0.89	p3, m3	2 / 4	**sbagliata**: "entrambe decadono verso zero"
	libs	0.94	m3	2 / 5	**corretta**, qualitativa ("laminare all'infinito")
Mistral Small 4	pipeline	0.89	r2, l1	4 / 5	generica; didascalia parafrasata, non quella del paper; nessuna formula
	single	0.78	p2, m3, l1	4 / 6	**due errori**: v "decade a zero" e φ "decade più lentamente"
	libs	0.89	m3, l1	3 / 6	**sbagliata**: "entrambi i profili decadono a zero"

Fatti delle rubriche: p2 = corrispondenza Reynolds ↔ costante di accoppiamento; p3 = laminare ↔ perturbativo; m3 = metodo di Heisenberg; r2 = parametri γ = 6; l1 = le equazioni di Navier-Stokes non sono equazioni di Eulero-Lagrange (limite dichiarato dall'autore). Per questo paper libs non ha estratto nessun dato: il grafico è vettoriale ma senza tacche numeriche leggibili sull'asse x, quindi non è stato calibrato. Le tre modalità "libs" hanno quindi ricevuto solo il testo.

Come ciascun modello descrive la figura (campo `figures.main_trend`)

Qwen3.8 · single: **corretto e quantitativo**.

> *"Curve 1 (v) starts at its maximum near 6 at x = 0 (approx., read from the figure), decays monotonically and levels off at a small nonzero value ≈ 0.5 at x ≈ 5 (approx., read from the figure), representing the turbulent core with v∞ ≠ 0. Curve 2 (φ) starts near 1 at x = 0 and decays to ≈ 0 by x ≈ 2 (approx., read from the figure), indicating that the flow becomes laminar at infinity."*

È la lettura migliore delle nove, ed è proprio quella che il controllo L2 penalizza: il valore "0.5" non è nel testo del paper, quindi risulta non ancorato (L2 0.969). È un esempio del perché il L2 di single su Qwen è leggermente più basso.

Qwen3.8 · pipeline: corretto, con le formule PySR aggiunte.

> *"Curve 1 shows a tube-like velocity profile and curve 2 decays toward zero at large x, corresponding to a laminar exterior as the fluctuation amplitude vanishes."*
> *visual_analysis_findings: curva 1 ≈ `6.24·exp(−0.588x)` (R^2 0.992), curva 2 ≈ `exp(−0.507x²)` (R^2 0.976)*

Le formule descrivono bene la discesa, ma `6.24·exp(−0.588x)` va a zero e non coglie il plateau a 0.6: PySR approssima la forma della curva nel tratto osservato, non la fisica. È giusto che siano marcate come "AI curve-fit approximation".

Gemma 4 · pipeline: generico nel testo, con formule simili a quelle di Qwen.

> *"Shows a strongly pronounced core (tube) for the velocity profile, suggesting a turbulent flow concentrated in a tube."*
> *visual_analysis_findings: curva 1 ≈ `9.95/(x² + 1.65)` (R^2 0.996), curva 2 ≈ `exp(−0.340x²)` (R^2 0.981)*

Gemma 4 · single: **errore di lettura**.

> *"Both v(x) and phi(x) peak at x = 0 and decay towards zero as x increases, indicating a localized core or tube-like structure."*

Gemma 4 · libs: corretto, riprendendo il testo del paper.

> *"The solution shows a strongly pronounced core (tube) for the velocity and a fluctuation profile that vanishes as x increases, indicating the flow becomes laminar at infinity."*

Mistral Small 4 · single: **due errori**, uno ripetuto anche nei risultati.

> *"The velocity profile v(x) shows a peak at the core (x ≈ 0) and decays to zero at large x, while the fluctuation function φ(x) has a smaller amplitude and decays more slowly, indicating laminar flow at infinity"*
> *key_results: "The z-component of velocity at infinity approaches zero (v→0 as ρ→∞), indicating the flow becomes laminar far from the core"*

La figura mostra un plateau a ≈0.6, e il testo indica $v_\infty \neq 0$ come caso più probabile ($v_\infty = 0$ è ammesso come "less probably"); φ decade più in fretta, non più lentamente. Questo errore però non pesa su nessuna metrica: la frase contiene comunque le parole chiave del fatto r1, e non cita numeri da verificare. È il limite delle rubriche a parole chiave, e un buon motivo per il giudice L3 ancora da fare.

Mistral Small 4 · pipeline: generico, didascalia inventata.

> *caption: "Profiles of velocity and correlation functions in turbulent flow" (quella vera: "The profiles of functions v(x) and φ(x), $\tilde{\alpha} = \beta = -3$, $\gamma = 6$ ...")*
> *main_trend: "The velocity profile shows a core region with higher velocity that decreases toward the edges, while the correlation function φ(x) describes the statistical fluctuations that peak at the center"*

In questo paper la visione di Mistral ha visto 3 curve invece di 2 e ha restituito punti segnaposto, che il sistema ha scartato. Per questo Mistral non ha ricevuto nessuna formula (nel job precedente, andato in OOM, era successa la stessa cosa). Perdendo la didascalia vera, perde anche il fatto r2 ($\gamma = 6$).

Mistral Small 4 · libs: **errore di lettura**.

> *"Both profiles exhibit a central peak with asymptotic decay to zero at large radii, indicating a tube-like structure with a core region of high activity surrounded by a laminar outer region"*

Il confronto è robusto tra i seed?

Il confronto sopra riguarda il seed 0. Ho ricontrollato la lettura della Fig. 1 anche negli altri due seed:

Output	Lettura corretta	Cosa dice su v(x) nei 3 seed
Qwen3.8 · single	3 / 3	sempre il plateau: "≈0.5" (s0), "roughly 0.5–0.6 by x ≈ 5" (s1), "decays toward a finite nonzero value v∞ … ~0.5" (s2); φ → 0 a x ≈ 2–3
Qwen3.8 · pipeline	1 / 3	v∞ ≠ 0 solo nel seed 0; negli altri la descrizione è qualitativa (core e φ → 0), ma senza errori
Gemma 4 · single	1 / 3	"both … decay toward zero" in s0 e s2; corretta in s1 ("phi(x) decays more rapidly")
Mistral Small 4 · single	0 / 3	v → 0 in tutti e tre i seed, sempre tra i `key_results` ("approaches zero", "→ 0", "approx. 0"); "φ decays more slowly" in s0 e s2
Mistral Small 4 · pipeline	–	generica in s0 e s1; v∞ → 0 tra i risultati in s2. I parametri α̃ = β̃ = −3, γ = 6 mancano in s0 e s2, ci sono in s1

Su questo paper, quindi, la superiorità di Qwen single nella lettura della figura è robusta, e l'errore di Mistral single è sistematico, non dovuto al caso. Resta però **un paper con una sola figura**: è un indizio sulla capacità visiva dei modelli, non una prova.

> **Un punto cieco delle metriche.** Sulle metriche aggregate (sezione 4) single è la modalità migliore proprio per Gemma e Mistral, eppure su questa figura è la loro peggiore. Le due cose sono vere insieme perché nessuna delle nostre metriche controlla se una figura è letta correttamente:
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> - la recall cerca parole chiave, e una frase come "v → 0 … laminar" le contiene;
> - il L2 controlla solo che i numeri citati compaiano nel testo, quindi penalizza Qwen single quando legge dal grafico un valore giusto ("≈0.5") e non vede l'errore di Mistral, che non cita numeri.
>
> Per una conclusione solida sulla visione serve un controllo specifico. Una possibilità è scrivere poche domande verificabili per ogni figura dei 10 paper (asintoto, quale curva decade prima, valore all'origine, numero di curve) e confrontarle con l'output. L'altra è il giudice L3.
> **Aggiornamento 29/10:** il controllo specifico ora esiste (sezioni 3 e 7). Sui controlli delle figure Qwen single fa 4/4 o 3/4 su Flux tube in tutti e 3 i seed, mentre Gemma e Mistral si fermano a 0–2 su 4.

Altre differenze visibili negli output

- **Ricchezza.** Qwen produce gli output più completi: circa 12–14 mila caratteri, 4–6 risultati con valore, condizione e fonte, 8 equazioni. Riporta anche il flusso cromoelettrico $\Phi_E = 2\pi\int f\, v\, \rho\, d\rho \neq 0$ e il suo analogo idrodinamico (Eq. 54). Gemma è la più sintetica (circa 6–7 mila caratteri, 2 risultati, 4–5 equazioni). Mistral sta in mezzo.
- **Limiti del paper.** Gemma (single e libs) e Qwen riportano il limite dichiarato dall'autore: le equazioni di Navier-Stokes non sono equazioni di Eulero-Lagrange. Mistral elenca invece 4 limiti generici, plausibili ma non scritti nel paper ("does not provide experimental or high-fidelity numerical validation"), e perde il fatto I1 in tutte e tre le modalità.
- **Caratteri corrotti in Mistral.** Alcuni simboli diventano `��` (per esempio "ρ��", "��α"). Sono caratteri UTF-8 multibyte spezzati nella decodifica dei token, causati da `skip_special_tokens: false`: confermato il 29/10 con un test sul server (20 caratteri rotti su una riga di simboli con `false`, nessuno con `true`, e il ragionamento resta separato anche con

`true`). Corretto negli script di Mistral. Non cambiano le metriche, ma per un dataset di fine-tuning andrebbero ripuliti.

- **Le modalità cambiano poco il riassunto.** I tre summary di ciascun modello dicono essenzialmente le stesse cose: analogia turbolenza–QCD, corrispondenza Reynolds ↔ accoppiamento, approssimazione a una funzione, soluzione a tubo cilindrica, flusso cromoelettrico non nullo. È la conferma, su un caso concreto, di quanto detto nella sezione 2: il contenuto viene dal testo, e la modalità incide solo sul modo in cui la figura viene descritta.

Appendice B: gli output JSON completi su Flux tube

Ecco i nove output completi, come li ha prodotti ciascun modello (sorgente PDF, seed 0), senza modifiche: compresi i caratteri corrotti di Mistral. Per ogni output indico le metriche e i fatti della rubrica mancanti. Ordine: per modello, poi pipeline → single → libs.

Qwen3.8 · pipeline

recall 0.94 (mancanti: m3) · L2 1.000 · 11,772 caratteri


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nonperturbative SU(3) gauge dynamics in quantum chromodynamics. It maps the Reynolds number to a dimensionless coupling
constant and models both problems by truncating the infinite hierarchy of velocity correlation functions or Green functions
using one or a few scalar order-parameter fields. In turbulence, this yields a cylindrically symmetric tube-like solution
of the Navier-Stokes equations with a core of turbulent flow surrounded by a laminar exterior. In QCD, the analogous
closure gives an effective SU(3) gauge theory containing scalar fields and a regular cylindrically symmetric solution with
a nonzero chromoelectric flux. The author argues this provides a possible hydrodynamic picture of a QCD flux tube, while
noting the analogy is not exact because Navier-Stokes equations are dissipative and not Euler equations of a Lagrangian.",
 "problem_addressed": "Whether a turbulent hydrodynamic system can contain a flux-tube-like solution analogous to the QCD
confinement flux tube, and how analogous closures of correlation or Green-function hierarchies can produce tube-like
solutions in both systems.",
 "motivation": "Turbulence and nonperturbative quantization of strongly interacting fields are both presented as major
unsolved problems involving fluctuating variables and infinite hierarchies of correlations or Green functions. The paper
seeks a connection between these problems and a possible QCD confinement mechanism, after discussing previous confinement
ideas such as dual superconductivity.",
 "main_contributions": [
  "Proposes the correspondence 1/\\tilde{g}^2 \\leftrightarrow \\rho v^2 l^4 and \\hbar \\leftrightarrow \\mu l^3, relating
the Reynolds number to a coupling constant and laminar flow to the perturbative regime and turbulent flow to the
nonperturbative regime.",
  "Derives a cylindrically symmetric tube-like turbulent solution of the incompressible Navier-Stokes equations from a one-
function approximation for velocity correlations.",
  "Derives an effective SU(3) gauge theory by decomposing QCD Green functions and obtains a regular cylindrically symmetric
flux tube filled with a color electric field with nonzero chromoelectric flux."
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assume an axially symmetric mean flow along z with zero-mean transverse fluctuations, neglect z and azimuthal-angle
dependence, and close second- and third-order velocity correlations using a single scalar function and constant
coefficients; then solve the resulting dimensionless ODEs for the mean longitudinal velocity and the fluctuation amplitude.
For QCD, decompose SU(N) gauge fields into a subgroup and a coset, use Heisenberg's nonperturbative expectation-value
equations, truncate the Green-function hierarchy by expressing four-point functions through two-point functions and two-
point functions through scalar triplet fields, obtain an effective Lagrangian and field equations, and solve them with a
cylindrically symmetric ansatz.",
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mean transverse velocity zero.",
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   "Third-order velocity correlation functions are expressed through second-order correlations with constant
coefficients.",
   "Derivatives with respect to z and the azimuthal angle are neglected in the turbulent transverse-velocity equation.",
   "In QCD, the subgroup SU(2) gauge fields are ordered with nonzero expectation values, while the coset SU(3)/SU(2) gauge
fields are disordered with zero expectation values.",
   "QCD four-point Green functions decompose into products or sums of two-point Green functions."
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   "Truncation of the infinite turbulent correlation hierarchy at third order.",
   "Truncation of the QCD Green-function hierarchy at fourth order by factorizing four-point functions.",
   "Neglect of z and azimuthal-angle derivative terms in the turbulent transverse-velocity closure.",
   "Use of numerical solution of the closed equations instead of an exact analytical solution."
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   "\\tilde{\\alpha} = \\frac{2}{3}\\alpha \\nu/\\varphi_0^2, \\tilde{\\beta} = \\frac{2}{3}\\beta/\\varphi_0",
   "The turbulent solution exists only for \\alpha < 0 and \\beta < 0.",
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  "Laminar incompressible flow, for which a cylindrically symmetric solution exists only in a pipe.",
  "SU(2) subgroup decomposition, which the paper says can lead to a Ginzburg-Landau Lagrangian, used as a related case."
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   "meaning": "Dimensionless closed equation for the mean longitudinal velocity in the turbulent tube solution."
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```

```
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Qwen3.8 · single

recall 1.00 (mancanti: –) · L2 0.970 · 13,617 caratteri


```
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theory)",
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quantization in quantum chromodynamics (QCD) are mathematically related, by comparing the Reynolds number with the
dimensionless coupling constant and proposing a correspondence 1/ḡ² ↔ ρv²l⁴ and ℏ ↔ μl³. Using a one-function approximation
for the velocity correlation functions (with the third-order correlation expressed through the second-order one), the
author closes the infinite hierarchy of correlation equations and derives a closed system of ordinary differential
equations for a cylindrically symmetric turbulent flow in unbounded space. The numerical solution is a tube-like profile
with a strongly pronounced velocity core that decays to a small nonzero value at infinity while the fluctuation function
decays to zero (laminar flow). Applying the same closure assumptions to the SU(3) Yang-Mills theory (Heisenberg's non-
perturbative method), the author obtains an effective Lagrangian whose field equations admit a regular cylindrically
symmetric solution carrying a nonzero color-electric field and a nonzero chromoelectric flux, interpreted as a QCD flux
tube. The author concludes that the two problems share a common structure (an infinite set of correlation/Green-function
equations that must be truncated) and that the turbulent core is an analog of the QCD flux tube stretched between a quark
and an antiquark.",
 "problem_addressed": "To find a cylindrically symmetric tube-like (flux-tube) solution in turbulent hydrodynamics as an
analog of the QCD flux tube, and to show that the same truncation of the infinite hierarchy of correlation/Green functions
that closes the turbulent equations also closes the QCD equations and yields a flux tube filled with a color electric
field.",
 "motivation": "Both the creation of turbulence near Re ≈ Recr and the nonperturbative quantization (quark confinement) for
α² ≥ 1 are listed as the last great unsolved problems of classical and quantum physics, respectively. Previous attempts to
explain confinement use the dual-superconductor/monopole-condensate scenario, nematic-liquid-crystal and two-band-dual-
superconductor pictures, and color-magnetic semi-superfluid vortices; the paper fills the gap of connecting these to
hydrodynamic turbulence through the common structure of the infinite correlation/Green-function hierarchy.",
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  "Proposes a correspondence between hydrodynamics and quantum field theory by identifying the Reynolds number with the
dimensionless coupling constant, via 1/ḡ² ↔ ρv²l⁴ and ℏ ↔ μl³ (Eqs. 4, 5).",
  "Obtains a cylindrically symmetric tube-like solution of the Navier-Stokes equations for a turbulent flow in unbounded
space by closing the correlation-function hierarchy with a one-function approximation (Eqs. 26, 27).",
  "Shows that the same closure assumptions applied to the SU(3) Yang-Mills theory produce an effective Lagrangian (Eq. 45)
whose field equations have a regular cylindrically symmetric solution with nonzero color electric fields and a nonzero
chromoelectric flux (Eqs. 52, 53), interpreted as a QCD flux tube."
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gauge fields) and a disordered fluctuating part (transversal velocity vρ, or the coset SU(3)/SU(2) gauge fields). The 2-
point and 3-point velocity correlation functions are expressed through a single scalar function φ(x) (one-function
approximation), and the 3-point correlation is expressed in terms of the 2-point one, which truncates the infinite
hierarchy. This yields a closed system of ODEs (Eqs. 26, 27) for the turbulent flow, solved numerically. In QCD the same
logic is applied to Heisenberg's operator equations: the SU(2) subgroup fields are taken in an ordered phase, the coset
fields in a disordered phase, the 2-point Green function is parameterized by real scalar triplet fields φᵃ (Eq. 43), and
the 4-point Green function is factorized into products of 2-point functions (Eq. 44), reducing the SU(3) Lagrangian to an
effective Lagrangian (Eq. 45) and a system of field equations (Eqs. 49-51) solved numerically with a cylindrical ansatz
(Eq. 48).",
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longitudinal fluctuations ⟨vₐ⟩ = 0",
   "The second- and third-order velocity correlation functions are expressed in terms of one function φ(xᵢ) (one-function
approximation)",
   "The third-order correlation function is expressed in terms of the second-order correlation function, allowing the
infinite set of correlation equations to be cut off",
   "In QCD the gauge field components Aᵃ_μ ∈ SU(2) are in an ordered phase with ⟨Aᵃ_μ⟩ = v̄ᵃ_μ (analog of v̄z)",
   "In QCD the gauge field components Aᵐ_μ ∈ SU(3)/SU(2) are in a disordered phase with ⟨Aᵐ_μ⟩ = 0 (analog of vρ)",
   "The 2-point Green function has the form G₂(x,y) = -η_μν f_mpₐ f_npᵇ φᵃ(x)φᵇ(y) and the 4-point Green function
factorizes into products of 2-point functions plus permutations (Eq. 44)"
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15)",
   "First-approximation description of the turbulent flow by fluctuating velocities v_{z,ρ} depending on ρ only",
   "Conjectural form of the 3-point correlation (Eq. 17) in which corrections to the leading term are restricted to the
quadratic compositions φ(yᵢ)φ(zᵢ) or φ(yᵢ)∂φ(zᵢ)/∂zᵢ",
   "Factorization of the fourth-order Green function as a product of second-order Green functions plus permutations (Eq.
44)",
   "Reduction of the SU(3) theory to an effective Lagrangian with parameters α, λ, m and (M²)^{μν} (Eq. 45)"
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turbulent flow in unbounded space",
   "Numerical investigation of the QCD field equations (Eqs. 49, 50, 51) with the cylindrical ansatz (Eq. 48) for the SU(3)
gauge theory"
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   "Existence condition for the turbulent tube solution: α < 0, β < 0",
   "QCD ansatz: A¹_t(ρ) = f(ρ)/g, A²_z(ρ) = v(ρ)/g, φ₃(ρ) = φ(ρ)/g, (M²)^{μν} = diag{M₀, M₁, 0, 0}",
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  "Previous QCD confinement scenarios cited in the introduction: dual superconductor / Abelian monopole condensation, SU(2) Yang-Mills as a nematic liquid crystal, two-band dual superconductor with Josephson coupling, and color-magnetic semi-superfluid vortices / BPS vortices"
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   "meaning": "Effective SU(3) Yang-Mills Lagrangian obtained after the Green-function closure; its field equations yield the QCD flux tube."
  },
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   "label": "Eqs. (49)-(51)",
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   "meaning": "The reduced ODE system (in the paper's notation with z denoting the radial derivative) for the QCD fields v,
f, φ under the cylindrical ansatz; its regular solution carries the flux tube."
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to about 5.",
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and levels off at a small nonzero value ≈ 0.5 at x ≈ 5 (approx., read from the figure), representing the turbulent core
with v∞ ≠ 0. Curve 2 (φ) starts near 1 at x = 0 and decays to ≈ 0 by x ≈ 2 (approx., read from the figure), indicating that
the flow becomes laminar at infinity. The authors read this as a tube-like (flux-tube) solution with a pronounced core."
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  "Inserting the turbulent flux tube into a laminar background flow (v∞ ≠ 0) is judged more probable than into a rest fluid
(v∞ = 0), so the rest-fluid case is less well supported.",
  "The author emphasizes that the analogy is not exact: any field theory has a Lagrangian whose Euler equations are the
field equations, whereas the Navier-Stokes equations are not Euler equations because of the presence of viscosity."
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correspondence and tube-like solutions as a starting point for relating the two unsolved problems."
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motivated by comparing the Reynolds number with a dimensionless coupling constant. It seeks a tube-like structure in a
turbulent fluid and a corresponding QCD flux tube by truncating the infinite hierarchy of correlation functions or Green
functions. Using a one-function correlation ansatz, the author obtains a cylindrically symmetric turbulent solution with a
core and plots v(x) and \\phi(x) for \\tilde{\\alpha}=\\tilde{\\beta}=-3 and \\gamma=6. Similar assumptions in SU(3) Yang-
Mills theory reduce the model to an effective Lagrangian whose ansatz equations have a regular solution with nonzero
chromoelectric flux. The paper concludes that turbulence and QCD share a common truncation structure, while noting that the
Navier-Stokes equations are not Euler equations of a Lagrangian because of viscosity.",
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can the infinite hierarchy of correlation or Green functions be truncated to produce a tube-like solution?",
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is associated with the nonperturbative regime of QCD. By comparing Reynolds number with a dimensionless coupling constant,
the paper argues that laminar flow corresponds to the perturbative regime and turbulent flow to the nonperturbative regime,
then uses hydrodynamics to motivate a QCD flux tube.",
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laminar/turbulent hydrodynamic regimes with classical, perturbative, and nonperturbative quantum-field regimes.",
  "Derives a cylindrically symmetric turbulent-flow tube solution from the Navier-Stokes equations using a one-function
correlation ansatz.",
  "Applies analogous Green-function truncation assumptions to SU(3) Yang-Mills theory and obtains a flux tube filled with a
color electric field.",
  "Presents a numerical turbulent-flow profile for \\tilde{\\alpha}=\\tilde{\\beta}=-3 and \\gamma=6 and shows that the
chromoelectric flux is nonzero."
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sources at z=\\pm\\infty, assumes axial symmetry and a mean flow along z, and splits the velocity into an ordered
longitudinal part and transverse fluctuations. The statistical hierarchy is closed by expressing second- and third-order
velocity correlations through one function \\phi(\\rho). The resulting dimensionless ODEs are solved numerically. In QCD,
the SU(3) gauge field is decomposed into SU(2) subgroup and SU(3)/SU(2) coset parts, field operators are introduced, and
expectation values generate a Green-function hierarchy. The hierarchy is closed by a 4-point function decomposition in
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   "In the transverse velocity equation, \\partial/\\partial z and \\partial/\\partial \\phi terms are neglected after
multiplying by v_\\rho and averaging.",
   "All second- and third-order velocity correlation functions are expressed through one function \\phi, and third-order
correlations are expressed through second-order ones with constants A_{abc}, \\alpha_a, and \\beta_a.",
   "In QCD, the SU(2) subgroup gauge fields are ordered with nonzero expectation values, while the SU(3)/SU(2) coset gauge
fields have zero expectation values.",
   "In QCD, the 2-point Green function is proportional to products of real scalar triplet fields \\phi^a, and the 4-point
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functions.",
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   "Turbulent solution condition: \\alpha < 0 and \\beta < 0.",
   "QCD case: SU(3) with G = SU(2), coset SU(3)/SU(2); SU(2) case: SU(2) with G = U(1).",
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turbulence, the dimensionless ODEs (26)-(27) are solved numerically and the profiles v(x) and \\phi(x) are plotted; for
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F^a_{\\mu\\nu}+\\frac{1}{2}(D_\\mu\\phi^a)(D^\\mu\\phi^a)-\\frac{\\alpha}{2}A^b_\\mu A^{c\\mu}\\phi^c\\phi^b-\\frac{\\
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regime of quantum chromodynamics (QCD). The author proposes a mathematical correspondence between the Reynolds number in
hydrodynamics and the dimensionless coupling constant in quantum field theory. By applying a 'one-function approximation'
to truncate the infinite hierarchy of correlation and Green functions in both systems, the author derives and numerically
solves equations that yield cylindrically symmetric tube-like solutions (flux tubes) in both turbulent flow and SU(3) gauge
theory.",
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nonperturbative quantization (specifically quark confinement) in QCD.",
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sets of coupled differential equations for correlation or Green functions, requiring similar truncation strategies for
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laminar flow to the perturbative regime and turbulent flow to the nonperturbative regime.",
  "Derived and numerically demonstrated a cylindrically symmetric tube-like solution for turbulent fluid flow using a one-
function approximation for correlation functions.",
  "Demonstrated that a similar approximation applied to SU(3) gauge theory leads to a flux tube solution with a nonzero
longitudinal color electric field and nonzero flux."
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   "Decomposition of the third-order correlation function in terms of the second-order one.",
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concentrated in a tube."
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recall 0.94 (mancanti: m3) · L2 1.000 · 5,984 caratteri


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chromodynamics (QCD), focusing on the existence of flux tube solutions in both systems. The author proposes that both
fields are characterized by infinite hierarchies of correlation or Green functions that require closure approximations to
be solved. By applying similar truncation methods, the author demonstrates that a cylindrically symmetric tube-like
solution exists for both a turbulent fluid and an SU(3) gauge theory filled with a color electric field.",
 "problem_addressed": "The difficulty of solving nonperturbative problems in both classical turbulence and quark
confinement in QCD, both of which involve infinite sets of coupled equations for correlation or Green functions.",
 "motivation": "The author seeks to find a connection between the 'legendary' unsolved problem of turbulence in classical
physics and the problem of color confinement in QCD, suggesting that the mathematical structure of their solutions
(specifically flux tubes) may be similar.",
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quantum field theory.",
  "Derived a cylindrically symmetric tube-like solution for turbulent flow using a one-function approximation for
correlation functions.",
  "Demonstrated that similar closure assumptions in SU(3) gauge theory lead to a flux tube filled with a color electric
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Eq. (17).",
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SU(3)/SU(2) coset are in a disordered phase."
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  "The author notes a fundamental difference: field theories have a Lagrangian from which field equations are derived,
whereas Navier-Stokes equations are not Euler equations due to the presence of viscosity."
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Gemma 4 · libs

recall 0.94 (mancanti: m3) · L2 1.000 · 6,732 caratteri


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quantum chromodynamics (QCD). The author proposes a correspondence between the Reynolds number in hydrodynamics and the
dimensionless coupling constant in quantum field theory. By applying a similar approximation method to cut off infinite
sets of coupled equations for correlation and Green functions, the author demonstrates the existence of cylindrically
symmetric tube-like solutions in both a turbulent fluid flow and an SU(3) gauge theory filled with a color electric
field.",
 "problem_addressed": "The author addresses the difficulty of understanding the creation of turbulence in classical physics
and the problem of nonperturbative quantization (specifically quark confinement) in quantum chromodynamics.",
 "motivation": "Both turbulence and nonperturbative QCD involve fluctuations—statistical in the former and quantum in the
latter—and result in an infinite set of coupled differential equations for correlation or Green functions that lack exact
analytical solutions.",
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(alpha^2) in quantum field theory.",
  "Derived a cylindrically symmetric tube-like solution for turbulent flow using a one-function approximation for
correlation functions.",
  "Demonstrated that similar assumptions in SU(3) gauge theory lead to a flux tube solution containing a nonzero color
electric field."
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functions in turbulence and Green functions in QCD. In turbulence, the longitudinal velocity is separated from disordered
transversal fluctuations. In QCD, the SU(3) gauge field is decomposed into an SU(2) subgroup (ordered phase) and an SU(3)/
SU(2) coset (disordered phase).",
  "key_assumptions": [
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   "For turbulence: third-order correlation functions are expressed as combinations of second-order correlation
functions.",
   "For QCD: the SU(2) subgroup components are in an ordered phase (expectation value equals the field), while coset
components are in a disordered phase (expectation value is zero).",
   "For QCD: 4-point Green functions are decomposed into products of 2-point Green functions."
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   "Use of a specific ansatz for gauge potentials and scalar fields to reduce the SU(3) field equations to ordinary
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   "Numerical investigation of field equations for an SU(3) gauge theory to find regular cylindrically symmetric
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For turbulence, the profiles of v(x) and phi(x) are computed. For QCD, the chromoelectric flux Phi_E is calculated to
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  "Dual superconductivity scenario (condensation of Abelian monopoles)",
  "BPS vortices in supersymmetric QCD",
  "Nematic liquid crystal interpretation of SU(2) Yang-Mills theory"
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A^b\\mu A^c\\mu \\phi_c \\phi_b - \\frac{\\lambda}{4}(\\phi_a\\phi_a - m^2)^2 - \\frac{1}{2}(M^2)^{\\mu\\nu} A^a\\mu A^a\
\nu",
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and the fluctuation function phi(x) (curve 2) on the vertical axis.",
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vanishes as x increases, indicating the flow becomes laminar at infinity."
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  "The Navier-Stokes equations are not Euler equations due to the presence of viscosity, which differs from the field
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 "code_and_data_availability": "Not specified",
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Mistral Small 4 · pipeline

recall 0.89 (mancanti: m3, r2) · L2 1.000 · 8,216 caratteri


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 "summary": "This paper proposes a mathematical analogy between turbulent fluid flow and quantum chromodynamics (QCD),
showing that both systems can support tube-like solutions. By assuming specific forms for correlation functions in
turbulent flow, the author derives a velocity profile characteristic of a flux tube. A similar analysis in SU(3) QCD yields
a color electric flux tube with nonzero chromoelectric flux. The work suggests that techniques from turbulence theory might
inform nonperturbative QCD calculations and vice versa.",
 "problem_addressed": "The paper addresses two fundamental unsolved problems: the nature of turbulence in fluid dynamics
and quark confinement in QCD, specifically seeking a theoretical framework that could provide insights into both phenomena
through their mathematical similarities.",
 "motivation": "The motivation is to bridge the gap between two major unsolved problems in physics by establishing a
theoretical connection between them. The author argues that both turbulence and nonperturbative QCD involve infinite
coupled equations for correlation/Green functions that cannot be solved exactly, and that solving one might inform the
other.",
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  "Establishes an explicit correspondence between Reynolds number in hydrodynamics and the coupling constant in QCD through
dimensionless parameters",
  "Derives a tube-like solution in turbulent flow under specific assumptions about velocity correlation functions",
  "Derives a similar flux tube solution in SU(3) QCD filled with color electric field, showing nonzero chromoelectric
flux",
  "Proposes that the mathematical structures underlying both systems are analogous, potentially enabling cross-
fertilization of solution techniques"
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infinite coupled equations. For turbulent flow, it assumes specific forms for velocity correlation functions to reduce the
infinite hierarchy of correlation equations to a solvable system. For QCD, it similarly decomposes gauge fields and assumes
forms for Green functions to derive effective field equations.",
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approximation)",
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   "The correlation function of the third order depends only on the correlation function of the second order",
   "Solutions exist only for negative values of certain parameters (α < 0, β < 0)"
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   "Decomposing fourth-order Green functions as products of second-order Green functions",
   "Using a single function to describe all correlation functions in turbulent flow",
   "Assuming cylindrical symmetry for the tube-like solutions"
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   "ρ0 = ν/φ0 (scaling parameter for turbulent flow)",
   "γ (parameter relating velocity components)"
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 "experiments": "The paper does not perform experiments but derives theoretical solutions and demonstrates that they
satisfy properties analogous to flux tubes in QCD, specifically showing nonzero flux in both turbulent flow and QCD
cases.",
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  "Classical laminar flow (for turbulent analogy)",
  "Perturbative regime in QCD (α² < 1) versus nonperturbative regime (α² ≥ 1)",
  "Previous approaches to quark confinement using dual superconductivity scenario and other models"
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\alpha} \\tilde{\\varphi} + \\tilde{\\beta} \\tilde{\\varphi}'\\quad v'' + \\frac{1}{x} v' = \\gamma \\left(\\tilde{\
\varphi} \\tilde{\\varphi}' + \\tilde{\\alpha} \\tilde{\\varphi} + \\tilde{\\beta} \\tilde{\\varphi}'\\right)",
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\varphi^2 + v^2 - M_0^2\\right)\\ \\varphi'' + \\frac{\\varphi'}{z} = \\varphi \\left(-f^2 + v^2 + \\lambda \\left(-\
\varphi^2 - m^2\\right)\\right)",
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and associated scalar field φ"
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functions of the radial coordinate x = ρ/ρ��",
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correlation function φ(x) describes the statistical fluctuations that peak at the center"
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   "The analogy between ⟨⟨ and viscosity is formal and not physically motivated beyond the mathematical correspondence",
   "The derived solutions are highly idealized and may not capture the full complexity of real turbulence or QCD"
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  "future_work": [
   "Investigate whether the analogy can lead to practical solution methods for either turbulence or nonperturbative QCD",
   "Explore more realistic correlation functions that might relax the one-function approximation",
   "Extend the analysis to more complex gauge groups or different fluid regimes",
   "Apply the derived flux tube solutions to specific physical scenarios in QCD"
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Mistral Small 4 · single

recall 0.83 (mancanti: p2, l1) · L2 1.000 · 9,422 caratteri


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quantum chromodynamics (QCD), proposing that both phenomena can be described by similar mathematical frameworks. The author
derives a set of reduced equations for a cylindrically symmetric turbulent flow and finds tube-like solutions that describe
a core with turbulent fluid surrounded by laminar flow. Extending these assumptions to QCD gauge theory, the paper
demonstrates the existence of a flux tube solution with non-zero color electric field, analogous to the flux tube between
quarks in QCD confinement.",
 "problem_addressed": "To find a mathematical and physical analogy between the unsolved problems of fluid turbulence and
quark confinement in QCD by developing a unified theoretical framework that describes tube-like structures in both
systems.",
 "motivation": "Both turbulence in fluid dynamics and quark confinement in QCD are long-standing unsolved problems in
physics. The author proposes that the mathematical structures underlying these phenomena might be similar, allowing
insights from one field to inform the other. This is motivated by the observation that both systems involve complex
fluctuations and the formation of tube-like structures under certain conditions.",
 "main_contributions": [
  "Derivation of a reduced set of equations describing a cylindrically symmetric turbulent flow in an unbounded space,
showing the existence of a tube-like solution with turbulent core and laminar exterior",
  "Introduction of a one-function approximation for correlation functions, allowing reduction of an infinite hierarchy of
equations to a closed system",
  "Extension of the turbulent flow model’s assumptions to quantum chromodynamics, demonstrating the existence of a flux
tube solution with non-zero color electric field in SU(3) gauge theory",
  "Establishment of a formal correspondence between the velocity field components in turbulent flow and gauge field
components in QCD"
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disordered components, then expressing correlation functions in terms of a single scalar function φ. This one-function
approximation allows reduction of an infinite hierarchy of correlation equations to a closed system. The same approach is
then applied to QCD by decomposing the gauge field and Green functions according to subgroup structure, leading to
effective field equations.",
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   "The flow is cylindrically symmetric and depends only on the radial coordinate ρ",
   "All degrees of freedom are split into ordered motion (longitudinal velocity component) and disordered fluctuations
(transversal velocity components)",
   "Correlation functions of second and third orders can be expressed in terms of a single function φ (one-function
approximation)",
   "The correlation function of the third order can be expressed in terms of the correlation function of the second order",
   "For QCD, the gauge field components are decomposed according to subgroup structure (SU(2) subgroup vs. coset SU(3)/
SU(2))"
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   "Neglect of certain terms in the Green function decomposition for QCD",
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   "Use of dimensionless variables to simplify the equations",
   "Cutoff of infinite hierarchy of equations using the one-function approximation"
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   "Negative values α < 0, β < 0 required for solution existence",
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   "Dimensionless variables x = ρ/ρ0, with ρ0 = ν/φ0",
   "Parameters ††α = (2/3)αν/φ0^2, ††β = (2/3)β/φ0, v = ††vz/φ0",
   "For QCD: specific ansatz A1^t(ρ) = f(ρ)/g, A2^z(ρ) = v(ρ)/g, φ3(ρ) = φ(ρ)/g"
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flow model and the effective QCD Lagrangian, analyzing the solutions for tube-like structures and non-zero flux, and
comparing the qualitative behavior of the solutions to known physical phenomena in both domains.",
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  "Previous approaches to QCD flux tubes and confinement mechanisms"
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   "meaning": "Expression for the non-zero color electric field components in the QCD flux tube solution"
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 "limitations": [
  "The one-function approximation truncates an infinite hierarchy of correlation equations, potentially missing important physical effects",
  "The model assumes cylindrical symmetry and neglects certain terms, which may not capture all aspects of real turbulent flow or QCD",
  "The connection between the two systems is formal and may not reflect deeper physical similarities",
  "The paper does not provide experimental or high-fidelity numerical validation of the proposed solutions"
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  "Extend the model to include more terms in the correlation function expansion to improve accuracy",
  "Investigate the stability of the tube-like solutions under perturbations",
  "Apply the approach to other gauge groups and compare with lattice QCD results",
  "Explore the phenomenological implications for quark confinement and hadron structure"
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 "code_and_data_availability": "Not specified",
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Mistral Small 4 · libs

recall 0.89 (mancanti: m3, l1) · L2 1.000 · 9,479 caratteri


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 "paper_type": "theoretical",
 "research_field": "physics / fluid dynamics and particle physics / quantum chromodynamics",
 "summary": "The paper proposes and explores a theoretical analogy between turbulent flow in fluids and flux tubes in
quantum chromodynamics (QCD). By assuming specific forms for velocity correlation functions in hydrodynamics, the author
derives a tube-like turbulent flow structure interpreted as an analog of QCD flux tubes responsible for quark confinement.
The approach is then extended to SU(3) gauge theory, yielding a cylindrically symmetric color electric flux tube with non-
zero chromoelectric flux, suggesting a shared mathematical structure between turbulence and nonperturbative QCD.",
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confinement in QCD, both of which involve complex, nonperturbative dynamics that resist exact analytical solution.",
 "motivation": "Both turbulence and nonperturbative QCD face significant theoretical challenges with no complete solutions
to date. The paper argues that since both systems involve fluctuating quantities (velocity in turbulence, gauge fields in
QCD) described by correlation functions, a comparative approach might reveal unifying principles or methods applicable to
both fields.",
 "main_contributions": [
  "Establishing a formal correspondence between the Reynolds number in hydrodynamics and the dimensionless coupling
constant in quantum field theory to map regimes between the two domains",
  "Deriving a cylindrically symmetric turbulent flux tube solution in hydrodynamics using a one-function approximation for
velocity correlations",
  "Demonstrating that similar assumptions in QCD (SU(3) gauge theory) lead to a flux tube filled with a color electric
field, with non-zero chromoelectric flux analogous to quark confinement"
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turbulence) or Green functions (in QCD) by expressing higher-order functions in terms of lower-order ones. The system is
split into ordered (mean) and disordered (fluctuating) components, and field or fluid equations are derived under
cylindrical symmetry assumptions.",
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   "Higher-order correlation/Green functions decompose into products of lower-order functions",
   "Gauge fields in QCD can be decomposed into an ordered SU(2) subgroup and a disordered coset SU(3)/SU(2)",
   "The turbulent flux tube exists in an unbounded space with sources at ±��, and the flow becomes laminar at infinity (v��
→ 0)"
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   "Neglecting azimuthal and axial derivatives (��/��z, ��/��φ) in the turbulent Navier–Stokes equations",
   "Truncating the infinite set of coupled equations by assuming specific forms for correlation functions to close the
system",
   "Approximating four-point Green functions as products of two-point functions in QCD",
   "Assuming the correlation functions are dominated by quadratic compositions of φ, neglecting higher-order corrections"
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profiles (Figure 1)"
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   "Dimensionless coupling constant α2 in QFT (α2 = 1/��g2)",
   "Parameters ��α, ��β, γ in the turbulent correlation model",
   "Parameters α, λ, m, and components of ��M 2 in the QCD effective Lagrangian",
   "Dimensionless radial coordinate x = ρ/ρ0 where ρ0 = ν/φ0"
  ]
 },
 "experiments": "The claims are substantiated by showing non-zero flux in both the turbulent flow model (Equation 54) and
the QCD flux tube model (Equation 53), with the existence of tube-like solutions under the stated assumptions. The
numerical integration of derived ODEs confirms the existence of cylindrically symmetric solutions.",
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  "Abelian monopole condensation scenarios for quark confinement (dual superconductivity)",
  "SU(2) Yang–Mills as a nematic liquid crystal",
  "Two-band dual superconductor models with interband Josephson coupling",
  "Color magnetic flux tubes in high-density QCD (semi-superfluid vortices)"
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   "source": "Fig. 1 and Sec. III"
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   "statement": "Non-zero chromoelectric flux ΦE through the QCD flux tube",
   "value": "ΦE = 2π ����^�� fv ρ dρ ��  0",
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   "source": "Eq. (53)"
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   "meaning": "Differential equation for the fluctuating velocity correlation function φ in the turbulent flow model under the one-function approximation"
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   "label": "Eq. (25)",
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   "meaning": "Differential equation for the mean longitudinal velocity component \\bar{v}_z in the turbulent flux tube model"
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   "meaning": "Effective Lagrangian for SU(3) gauge theory under the proposed decomposition and one-function approximation, leading to the flux tube solution"
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   "meaning": "Components of the color electric field in the QCD flux tube solution, where f and v are functions of the radial coordinate"
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   "meaning": "Non-zero chromoelectric flux through the QCD flux tube, analogous to the flux in the turbulent flow core"
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   "caption": "Profiles of the mean velocity v(x) and the correlation function φ(x) in the turbulent flux tube model",
   "what_it_shows": "The radial profiles of the dimensionless mean velocity v(x) (curve 1) and the correlation function φ(x) (curve 2) as functions of the dimensionless radius x, for parameters ��α = ��β = -3 and γ = 6. The velocity profile shows a pronounced core with decay to zero at infinity, while the correlation function peaks near the center and decays.",
   "main_trend": "Both profiles exhibit a central peak with asymptotic decay to zero at large radii, indicating a tube-like structure with a core region of high activity surrounded by a laminar outer region"
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  "The model relies on heuristic assumptions about correlation functions that may not capture the full complexity of real turbulent flow",
  "Solutions are obtained numerically and are not exact analytical solutions",
  "The analogy between turbulence and QCD is formal and may not reflect physical equivalence beyond the mathematical structure",
  "The one-function approximation is a strong simplification that neglects many potential higher-order correlations"
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  "Investigating whether the proposed analogy can lead to analytical solutions in either turbulence or QCD",
  "Exploring experimental or lattice QCD tests to validate the existence of flux tube solutions analogous to the turbulent model",
  "Extending the model to incorporate more realistic correlation functions or additional degrees of freedom",
  "Applying similar truncation methods to other nonperturbative problems in physics"
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Dati: `logs_cineca/matrix/*.jsonl` (output), `logs_cineca/matrix/matrix_eval_local.txt` e `matrix_report_local.jsonl` (valutazione), `gold/rubrics.jsonl` (rubriche).